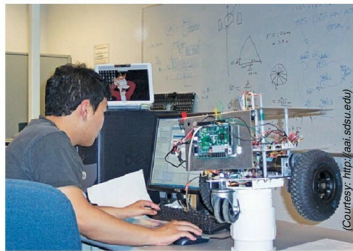


FPGA is an Evergreen, Niche, Not-Easy-To-Get-A-Job Industry

Take a look at the opportunities, scope, perks and skills expected from an engineer looking to work in the FPGA domain, topped with some advice from industry experts in this field



(Courtesy: <http://aii.sdsu.edu>)

ABHISHEK A. MUTHA

An increased adoption of field-programmable gate arrays (FPGAs) in the semiconductor industry has opened tremendous opportunities for FPGA engineers. Current usage of FPGA varies from a simple interface logic implementation to FPGA as the main processing device in the system.

Paramita Kapat, head—human resource, eInfochips adds that, “With increased resource size and powerful DSP blocks, processor blocks, transceiver blocks and large on-chip memory, FPGAs play a major role in a variety of domains including video processing, networking, avionics, ATE and as hardware accelerators.” This has increased the demand for competent FPGA engineers in the industry.

Growth of FPGA usage driving career scope

FPGAs are widely being used in various industries ranging from medi-

cal applications and handheld devices to aerospace and military segments. “Hence the FPGA industry offers an immense potential for both the industry and for aspiring engineers to work, learn and grow,” notes Rajesh Chakkingal, lead architect, Mistral Solutions.

On a similar note,

Rajesh Choudhary, HR head, Xilinx India informs, “FPGA as a technology is evolving and has become predominant in various domains such as telecommunication infrastructure, industrial automation, aerospace and high-end consumer electronics such as 3D TVs and personal digital assistants. The career scope of FPGAs is also increasing with their increased usage.”

These days, there are various career choices available to engineering graduates as well as postgraduates. With over 20 years of industrial experience, Yogesh Soni, senior technical director, Mattozetta Technologies believes a

career in very-large-scale integration (VLSI) is one of the best choices available today.

Although, he says, “It is tough for a fresher to enter into VLSI as a career, but once you are in, you would have a world-class career. Within VLSI, a career in FPGA is an evergreen choice.” As technology nodes have shrunk to very low level, thereby skyrocketing the application-specific integrated circuit (ASIC) development cost, FPGA provides a very good alternative to quick and faster design, and development for any product.

Multitude of roles

The two broad opportunities in this field are with designing FPGA-based designs and then verification. Both these areas are equally challenging, and require high level of involvement. Chakkingal adds that, “At the entry level, engineers are expected to learn hardware description languages (HDLs) such as VHDL/Verilog and use them for register-transfer level (RTL) coding and verification of test benches.”

Also, entry-level engineers can target jobs in logic design/verification in the FPGA application areas. Choud-

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